Assignment 3: PCA and Model Comparison

Project Overview

This project explores dimensionality reduction using Principal Component Analysis (PCA) and compares multiple machine learning classification algorithms using the Wine Quality dataset. The objective is to preprocess the dataset, reduce the number of features, train classification models, and identify the best-performing model using suitable evaluation metrics.

Objectives

- Load and clean the Wine Quality dataset.
- Apply Standard Scaling to numerical features.
- Perform Principal Component Analysis (PCA).
- Select principal components based on explained variance.
- Train Logistic Regression, K-Nearest Neighbors (KNN), and Gaussian Naive Bayes classifiers.
- Compare model performance using classification metrics.
- Identify the best-performing model.

Dataset

Dataset: Wine Quality – Red Wine

Source: UCI Machine Learning Repository

The dataset contains physicochemical properties of red wine and a quality score.

- **Input features:** 11 numerical features.
- **Target variable:** quality.
- **Original dataset size:** 1,599 records.
- **Classification type:** Multiclass classification.

Technologies Used

- Python
- Google Colab
- Pandas
- NumPy

- Matplotlib
- Seaborn
- Scikit-learn

Project Workflow

1. **Data Loading:** Load the Wine Quality dataset.
2. **Data Cleaning:** Check for missing values and duplicate records.
3. **Feature Selection:** Separate the input features and target variable.
4. **Data Splitting:** Divide the dataset into training and testing sets.
5. **Standard Scaling:** Standardize the input features.
6. **PCA:** Reduce dimensionality while retaining the selected percentage of explained variance.
7. **Model Training:** Train three classification algorithms.
8. **Model Evaluation:** Compare accuracy, weighted precision, weighted recall, and weighted F1 score.
9. **Visualization:** Generate explained variance plots, model comparison charts, and confusion matrices.
10. **Best Model Selection:** Select the model based on its weighted F1 score.

Machine Learning Algorithms

1. Logistic Regression

A classification algorithm used to predict wine quality classes based on the transformed features.

2. K-Nearest Neighbors (KNN)

A distance-based classification algorithm that predicts a class using the labels of nearby training samples.

3. Gaussian Naive Bayes

A probabilistic classification algorithm based on Bayes' theorem, assuming conditional independence of features and Gaussian feature distributions within each class.

Evaluation Metrics

- **Accuracy:** Proportion of correct predictions.
- **Precision:** Proportion of predicted class instances that are correct.

- **Recall:** Proportion of actual class instances correctly identified.
- **Weighted F1 Score:** Weighted harmonic mean of precision and recall across classes.
- **Confusion Matrix:** Shows the relationship between actual and predicted classes.

PCA Analysis

PCA transforms the original features into principal components ordered by explained variance. The number of components is selected based on the cumulative explained variance, targeting at least 95%.

The selected components retain most of the variation in the standardized training features while reducing dimensionality. Each component is a weighted combination of the original features.

Results and Model Comparison

The three models are evaluated using the same training/testing split and PCA-transformed features.

Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1 Score
Logistic Regression	From notebook	From notebook	From notebook	From notebook
K-Nearest Neighbors	From notebook	From notebook	From notebook	From notebook
Gaussian Naive Bayes	From notebook	From notebook	From notebook	From notebook

Best-performing model: To be determined from the actual experiment results.

The model with the highest weighted F1 score is selected as the best-performing model for this experiment. Replace the placeholders in the table with the actual values obtained from Google Colab.

Visualizations

The project generates the following visualizations:

- PCA cumulative explained variance plot.
- Model performance comparison chart.

- Classification reports for all three models.
- Confusion matrix for the selected model.

How to Run the Project

1. Open Google Colab.
2. Upload or open Assignment_3_PCA_Model_Comparison.ipynb.
3. Run all notebook cells in order.
4. Review the PCA analysis and evaluation metrics.
5. Compare the models and identify the best-performing algorithm.

Conclusion

This project demonstrates how PCA can reduce the dimensionality of a dataset and how multiple classification algorithms can be compared using consistent evaluation metrics. The results help identify the model that performs best on the Wine Quality classification task according to the chosen evaluation criterion.

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